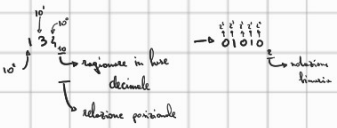
$y = f(x) \quad y \in \mathbb{B} \quad x \in \mathbb{B}^n$

$x \in \mathbb{B}$ bit





$N = b_n \ldots b_0 \quad b_i \in \mathbb{B}$

$= b_n B^n + b_{n-1} B^{n-1} + \ldots + b_0 B^0$



n bit

$\mathbb{N}^+ = \{0, 1, \ldots\}$

$\mathbb{N}_{nat} = \{0, \ldots, 2^n - 1\}$

$12_{10} = 1 \cdot 10^1 + 2 \cdot 10^0 = 10 + 2 = 12$

$0101_2 = 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 + 0 \cdot 2^0 = 10_{10}$

x_3	x_2	x_1	x_0	dec
0	0	0	0	0
0	0	0	1	2
0	0	1	1	3
...				...
1	1	1	1	15

$10\,100\,11\,00\,10_2$

$= 2 + 16 + 32 + 256 + 1024 = 1____{10}$

$1001_2 \to 2^0 + 2^3 = 9_{10}$

100110

38	0
19	1
9	1
4	0
2	0
1	1
0	

9	1
4	0
2	0
1	1
0	

LSB ... MSB

basi:

- **ottale** (oct) $\to \{0, \ldots, 7\}$

$33_8 = 3 \cdot 8^1 + 3 \cdot 8^0 = 24 + 3 = 27_{10}$

27	3
3	3
0	

$3_{10} \to 3_8 \to 011\,011$

$011\,011 \to 1 + 2 + 8 + 16 = 27_{10}$

X_2	X_1	X_0	
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

$001\,010 \to 12_8$

$1 \quad 12$

- esadecimale (HEX)

x^3	x^2	x^1	x^0	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A (-10)
1	0	1	1	B (-11)
1	1	0	0	C (-12)
1	1	0	1	D (-13)
1	1	1	0	E (-14)
1	1	1	1	F (-15)

$0101 \; 1110_2 \to 5E_{16}$

5 E

$5 \cdot 16^1 + E \cdot 16^0 = 5 \cdot 16 + 14 \cdot 16^0$

$80 + 14 = 94_{10}$

94	0
47	1
23	1
11	1
5	1
2	0
1	

94	14
5	5
0	

$\to 5E_{16}$

BCD 8421

DEC	
0	0000
1	0001
2	
3	
4	
5	
6	
7	
8	
9	1001

1013

0001 - 0000 - 0001 - 0011

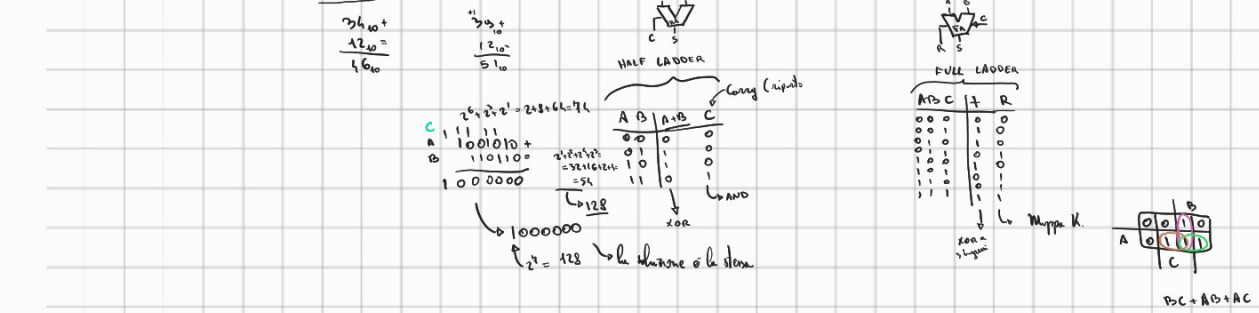
altri codici sono $\to$ i codici GRAY

1 bit	2 bit	3 bit	4 bit
0	00	000	0 000
1	01	001	0 001
	11	011	0 011
	10	010	0 010
		110	0 110
		111	0 111
		101	0 101
		100	1 100
			1 101
			1 111
			1 110
			1 010
			1 011
			1 001
			1 000

espansione della mappa di Karnaugh

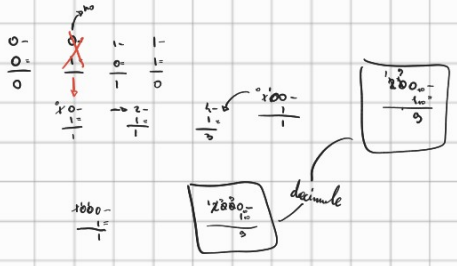
Somme:

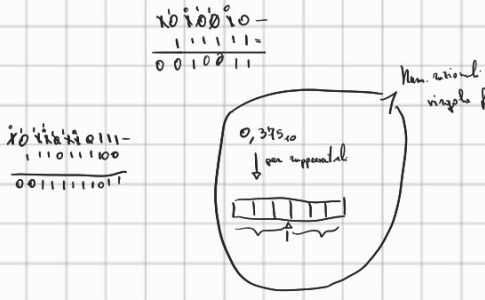
half adder



Sottrazioni:

di 1 bit



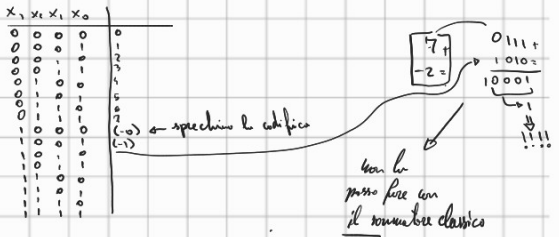


Prodotto



Numeri negativi

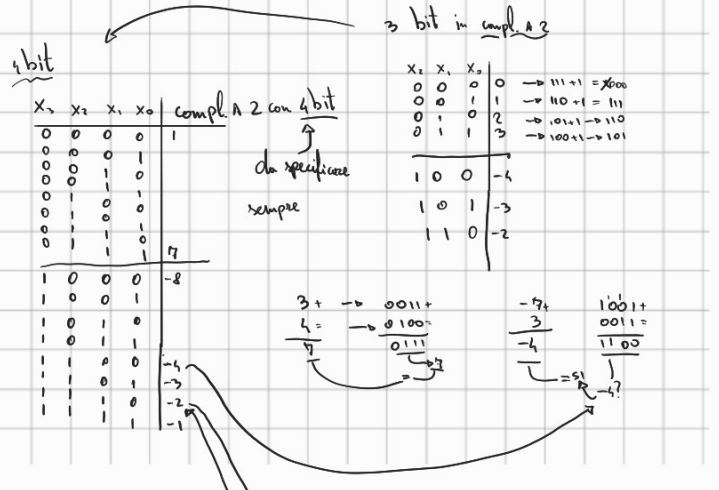
(rappresentazione complementi)



Usiamo il metodo del COMPLEMENTO A 2 $[-2^{n-1}, \dots, 0, \dots, 2^{n-1}]$

↳ X → NEGO TUTTO → SOMMO 1

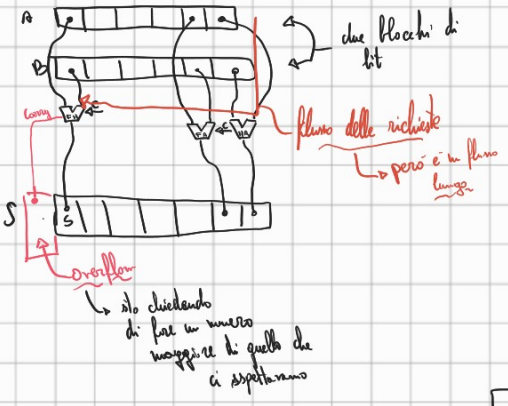
in più positivi $[0, \dots, 2^{n-1}-1]$

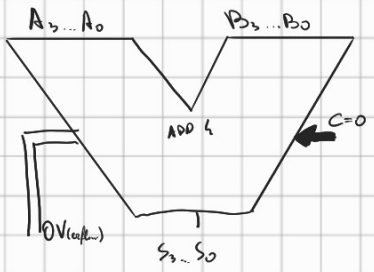


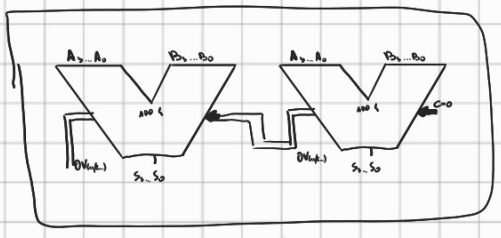


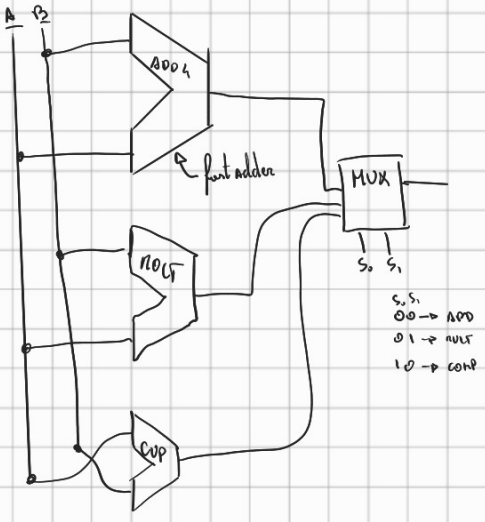


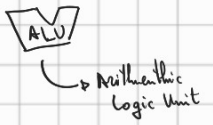
SOMMATORE A CASCATA













Dove m'ero chiusta al libro

B

B

B

B
PAR

Spiegazione dell'es. 1.6

↳ vediamo se voglio trasmettere dei bit



